UNREAL ENGINE PLUGIN

Minimap & Map System

A HUD minimap and a full-screen map that share one set of markers, one coordinate system and one set of icons. Put a component on an actor and it is on the map.

Unreal Engine 5.6, 5.7, 5.8 · **Setup time** about five minutes · **Blueprint** everything in this guide

1 What you are looking at

> **INSTALLING**
>
> In the Epic Games Launcher, go to Library → Fab Library, find **ACGA Minimap & Map System**, choose **Install to Engine** and pick your engine version. It arrives already compiled, so a Blueprint-only project works as it is.
>
> In your project, open Edit → Plugins, search for *ACGA Minimap*, tick it and restart the editor. To see its assets, open the Content Browser's Settings and tick **Show Engine Content** and **Show Plugin Content**.

> **PRESS PLAY BEFORE YOU READ THE REST**
>
> The plugin ships a finished, playable level. Open `ACGAMinimap → MinimapSystem → Demo → Maps → DemoMap_ACGA_Minimap` and press Play. There is already a character on it, with the controls set up and the level told to use it, so nothing in your project needs configuring first — and nothing in your project is changed.
>
> It is a walled district with five landmarks and a road grid. WASD and the mouse move and look. **M** opens and closes the big map; click a marker on it to select it. **N** switches the minimap off and back on while you play.
>
> Two things are worth trying straight away. **Turn on the spot** and watch the minimap: whether it turns with you or stays fixed is a single setting, and it is the first one most people want to change. Then **select several markers** on the big map and watch the oldest one let go once you pass the limit.
>
> Last, **load it up**: open the console with the tilde key and type `ACGA.Minimap.Demo.Bench show 2000` . The big map opens with two thousand markers moving across it and a readout of what they cost each frame. Add `widgets` at the end to see the same markers drawn as one UMG widget each, and type `ACGA.Minimap.Demo.Bench stop` to end it. Section 9 has the measured numbers.

This guide is for whoever places things in the level and decides how they look. You will not need to write any code, and nothing here asks you to open a C++ file.

The system has four pieces. You will only ever place one of them by hand.

PIECE	WHAT IT IS FOR
ACGA Map Capture	An actor you drop into the level once. Its box says how much of the world the map covers, and it photographs that area from above to make the map picture.
ACGA Map View	A widget you drag into any UMG screen. A minimap is this with a circle and a small zoom; the full map is the same thing showing the whole world.
ACGA Map Marker	A component you add to any actor to put it on the map.
Map Icon Set	A data asset holding the icons. This is where you add your own art.

> **THE ONE IDEA WORTH KNOWING**
>
> The Map Capture's **box is the map**. Its size decides what area the picture covers and it decides where every marker lands. There is no second place to set the world size, so the picture and the markers can never drift apart.

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4 Your own icons

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8 Hiding the map

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2 Getting it on screen

Five minutes, start to finish. Do these in order.

1. **Turn the plugin on.**
   Edit → Plugins, search for *ACGA Minimap*, tick it, restart the editor when it asks.

2. **Drop the capture actor into your level.**
   In the Place Actors panel, search for *ACGA Map Capture* and drag it in. Put it roughly in the middle of your playable area — the exact position does not matter yet.

3. **Stretch its box over the playable area.**
   Select it, find `Bounds Box` in the Details panel and scale it until the blue box covers everywhere the player can go. Look at it from the top: only the width and depth matter, the height just has to contain the buildings you want in the picture.

4. **Make a HUD widget, or open the one you have.**
   In the UMG palette, find
   ACGA Map View
   under the *ACGA* category and drag it onto your canvas. Drag it to a corner and size it — 240 × 240 is a good starting minimap.

5. **Set how much world it shows.**
   With the map view selected, set `Zoom World Radius`. This is in centimetres, measured out from the player. 3000 is close in, 8000 is a wide view. Unreal's default character is about 180 cm tall, if you need something to judge it against.

6. **Show the widget.**
   Add it to the screen however your project already does — usually *Create Widget* and *Add to Viewport* in your player Blueprint.

7. **Press Play.**
   You should see the map picture, a circular frame, and a blue arrow in the middle that turns as you turn.

> **IF THE MAP IS BLANK**
>
> The capture takes its picture when the level starts. If your level was still loading textures at that moment, the picture can come out grey. Select the capture actor and press the **Bake Background** button in its Details panel to take a fresh one from the editor.

3 Putting things on the map

1. **Select the actor**
   you want on the map — a chest, a door, an objective, a vehicle.

2. **Add Component → ACGA Map Marker.**

3. **Pick its icon**
   from the `Icon Id` dropdown.

That is the whole thing. There is nothing to register and nothing to remember to clean up — destroying the actor takes it off the map.

The settings you will actually touch

SETTING	WHAT IT DOES
Icon Id	Which icon, from a dropdown of what is in the icon set.
Tint	Colours the icon. Use it to tell two of the same icon apart — a friendly and an enemy flag.
Scale	Makes this one marker bigger or smaller than the rest.
Update Policy	How often the map re-reads where it is. See the note below — this is the one that matters for performance.
Rotate With Actor	Turns the icon to face where the actor faces. On for anything that moves and has a front, off for pins and objectives.
Category	A label you invent. A map view can then show only certain categories — quest markers on one map, resources on another.
Start Visible	Untick for something that appears later, when the player discovers it.
Selectable	Lets the player click it on the map. See section 7.

> **UPDATE POLICY, IN PLAIN TERMS**
>
> **Static** — for anything that never moves. Never re-read once placed; only drawing it costs anything, so you can have thousands.
> **On Move** — the default. Checks whether it moved, and only does work if it did.
> **Every Frame** — for the player, teammates, vehicles. Use it where you need it and not elsewhere.

4 Your own icons

This is the part you will come back to. Icons live in one asset, and adding yours takes about a minute.

The plugin ships with a set at:

ACGAMinimap ▸ MinimapSystem ▸ Icons ▸ DA_ACGA_MapIcons_Default

It contains `ViewTarget` (the player arrow), `Player`, `Objective`, `Question`, `Alert`, `Dot` and `Default`.

1. **Duplicate it**
   into your own Content folder. Right-click → Duplicate. Work on the copy, so that updating the plugin later cannot overwrite your work.

2. **Add a row**
   to the Icons list.

3. **Give it a name and a texture.**
   The name is what you will pick from the dropdown, so make it something you will recognise — `Campfire`, not `Icon_04`.

4. **Point the project at your copy.**
   Edit → Project Settings → Plugins → ACGA Minimap → `Default Icon Set`.

Your name now appears in the `Icon Id` dropdown on every marker.

Per-icon settings

SETTING	WHAT IT DOES
Icon Id	The name you will pick in the dropdown.
Texture	Your art. Square works best. A white shape you tint per marker is more flexible than pre-coloured art.
Clamped Texture	Drawn instead when the thing is off the edge of the map — normally an arrow. Draw it pointing **up**.
Size	How big it is drawn, in screen units.
Tint	A colour baked into this icon. The marker's own tint multiplies on top.
Draw Priority	Higher numbers draw on top. Put the player above everything else.

> **ABOUT TEXTURE SETTINGS**
>
> Import your icons as normal textures. The shipped ones use the **UI** texture group with compression set to **UserInterface2D**, which keeps hard edges crisp. If yours look muddy, that is the setting to copy.

> **SAME NAMES IN BOTH PLUGINS**
>
> The ACGA Screen Waypoint plugin has an icon set built the same way, with the same starting names. If you use both, you can say `Question` once and mean it in both places. They are separate assets so that either plugin can be removed without breaking the other.

5 Styling the map view

Set these once for the whole game, not once per widget.

Every setting in this section exists in two places: on the **ACGA Map View** widget, in *View Settings*, and in **Project Settings → Plugins → ACGA Minimap**, in two blocks — *Minimap Defaults* and *Full Map Defaults*.

The Project Settings blocks are the ones in use. Every map view ships with *Use Project Defaults* ticked, and a ticked view reads the block that matches its *Role*: a Minimap-role view reads Minimap Defaults, a Full Map-role view reads Full Map Defaults. Change a value there and every map in the game changes with it.

> **THIS IS HOW YOU RESTYLE THE FULL MAP**
>
> You never place the full map yourself — pressing your map key builds it for you — so *Full Map Defaults* is the only way to change how it looks without replacing the whole widget. Border colour, whether it fills the screen, how big the icons are: all of it is in that block.

To make one particular view disagree with the rest of the game, untick *Use Project Defaults* on it and fill in its own values. A view whose Role is **Custom** ignores the project blocks entirely and always uses its own — that is what Custom is for.

SETTING	WHAT IT DOES
Use Project Defaults	Take everything below from Project Settings instead of from this widget. **On by default.**
Role	**Minimap**, **Full Map** or **Custom**. This decides which of the two "hide the map" switches this view listens to — see section 8. Use **Custom** for a map on a wall panel or a table that should never be hidden by a HUD toggle.
Shape	Circle or rectangle. The circle is a real circular edge, not a square with a mask image — you do not need to supply any art for it.
Rotation	North Up, Follow View Target, or **Use Project Default** (the default). See the note below.
Zoom World Radius	How many centimetres out from the centre the view shows. **0 means the whole world**, which is what a full map wants.
Follow View Target	Keeps the player in the middle. Turn it off and the view stays fixed on the centre of the world.
Clamp Mode	What happens to markers outside the view: not drawn, pinned to the edge, or pinned and swapped for their arrow art pointing the right way.
Clamp Padding	How far in from the edge those pinned markers sit, so they do not sit on the frame.
Icon Scale	Scales every icon in this view. A minimap usually wants less than 1.
Draw View Target Icon	The player arrow in the middle.
Category Filter	Leave empty to show everything. Fill it in to make a view that only shows some kinds of marker.
Void Color	What fills the view where the map does not reach — past the edge of the world.
Draw Border	The outline. Colour and thickness next to it.
Circle Segments	How smooth a circular edge is. 64 is smooth at any HUD size.
Selection colours	The ring drawn around a selected marker — colour, thickness and the gap to the icon.

WHETHER THE MINIMAP TURNS

Set it in **Minimap Defaults** → **Rotation** and every minimap in the game follows, which is what lets a single "rotate minimap" option in your settings menu change all of them at once. There is a matching Blueprint node, **Set Minimap Rotation**, for exactly that option.

The **Use Project Default** value you see on the widget itself means the same thing one level down: it is there for a view that has *Use Project Defaults* unticked but still wants to follow the game-wide answer on this one setting.

6 The full map

Bind a key to **Toggle Full Map**. In a Blueprint that is one node — search for it by name; you do not need to get anything first.

It opens a full-screen map, shows the mouse cursor, and puts the cursor and the input mode back exactly as they were when it closes.

The map it opens is the plugin's own, which works out of the box. To use your own design instead: make a UMG widget, drag an **ACGA Map View** into it, set its `Role` to **Full Map** and its `Zoom World Radius` to 0, then point Project Settings → Plugins → ACGA Minimap → `Full Map Widget Class` at it.

> **TURNING IT OFF**
>
> If your game has no full map, untick `Full Map Enabled` in Project Settings. The open node then does nothing, so nobody has to remember not to bind the key.

7 Letting the player select things

Tick `Selectable` on a marker, and `Allow Selection` on the map view. Clicking the marker selects it; clicking it again unselects it. Selected markers get a coloured ring.

How many at once

Project Settings → Plugins → ACGA Minimap → `Max Selected Markers`. Set **0** for no limit. Otherwise, when the player picks one more than the limit, the one they picked **longest ago** is dropped — so "you may track up to three objectives" needs no work from you beyond typing 3.

Reacting to it

The map marker component has three events you can use in its Blueprint:

EVENT	FIRES WHEN
On Marker Selected	The player picks it.
On Marker Unselected	They unpick it, the limit pushed it out, or the actor went away while selected.
On Marker Selection Changed	Both of the above, with a true/false, if you want to handle them together.

> **A COMMON THING TO BUILD**
>
> Use *On Marker Selection Changed* to switch a world waypoint on and off, so picking a place on the map puts a marker on screen pointing at it.

8 Hiding the map

Two switches, each with a project setting for how the game starts and a Blueprint node for changing it during play.

	PROJECT SETTING	BLUEPRINT NODE
Minimap	Minimap Enabled	Set Minimap Enabled
Full map	Full Map Enabled	Set Full Map Enabled

Set Minimap Enabled (false) takes every minimap off screen at once, wherever it sits in your HUD — so a photo mode or a cutscene is one node, not a hunt for the widget that happens to contain the map. Views set to **Custom** are never affected.

Set Full Map Enabled (false) also closes the map if it is open, so locking the map during a chase does not also need you to block the key.

9 Every setting, in one place

Project Settings → Plugins → ACGA Minimap

SETTING	WHAT IT DOES
Default Icon Set	The icon library the whole project uses.
Snapshot Hz	How often moving markers are re-read. 30 looks the same as 240 and costs far less.
Move Threshold	How far something must move, in cm, before the map bothers updating it.
Initial Marker Capacity	How many markers to make room for up front. Set it to roughly your real number.
Poll Marker Interface	Leave on unless you have thousands of markers and need the last bit of performance.
Selection Radius	How close a click has to be to count as picking a marker.
Max Selected Markers	How many can be selected at once. 0 is no limit.
Minimap Defaults	How every minimap-role view looks: shape, rotation, zoom, clamping, icon scale, colours, border, selection ring. Read by every view with *Use Project Defaults* on, which is all of them until you say otherwise.
Full Map Defaults	The same block for the full map. This is the only way to restyle the full map without replacing the widget, because you never place that widget yourself.
Minimap Enabled	Whether the minimap is on screen when a level starts.
Full Map Enabled	Whether the full map can be opened at all.
Full Map Widget Class	Your own full-map screen. Empty uses the one that ships.
Full Map Z Order	What the full map draws in front of.
Full Map Manages Input Mode	Leave on unless your game handles cursor and input modes itself.
Debug Draw	Prints what the map thinks is going on, over the game. See section 10.

On the ACGA Map Capture actor

SETTING	WHAT IT DOES
Bounds Box	**The map.** Scale it over the playable area.
Background Source	**Baked Capture** — one picture when the level starts, the normal choice. **Live Capture** — re-photographed every frame, for worlds that change shape. **Supplied Texture** — your own hand-painted map.
Supplied Texture	Your hand-painted map, for the mode above.
Capture Resolution	Resolution of the longer side. The other side follows your box's shape automatically.
Flatten Capture For Map	Takes the picture without fog, bloom or vignette. Leave on: a map is a diagram, not a screenshot.
Capture Exposure Bias	Brightness of the map picture, in stops. Negative is darker. Adjust this if the baked map looks washed out or too dark.
Rebake Delay	Takes a second picture a moment after the level starts, in case the first caught the level still loading. Leave it alone.
Capture Height	How high the camera sits. It has to be above your tallest building.
Bake Background	A button. Takes the picture now, from the editor.

How much markers cost

Markers are not widgets. The map keeps them in one list and draws them all in one pass, so a marker costs far less than an icon placed in UMG. The demo includes the test these numbers come from: it moves markers across the full map every frame, first with this plugin, then as one UMG image per marker — the usual way a map is built in UMG, written the cheapest way UMG allows.

MARKERS	THIS PLUGIN	ONE UMG WIDGET PER MARKER
250	0.04 ms	0.60 ms
500	0.07 ms	0.99 ms
1 000	0.18 ms	1.89 ms
2 000	0.34 ms	3.76 ms
4 000	0.73 ms	7.25 ms

Time per frame spent drawing and moving the markers, measured at 1920 × 1080 in Unreal Engine 5.6 on an Intel Core i5-13400F with a GeForce RTX 5070. To measure your own machine, open the demo map, press Play and type `ACGA.Minimap.Demo.Bench sweep` in the console. It takes about a minute and saves the results in your project's `Saved/ACGA_MinimapBench` folder.

10 When something does not look right

Start by turning on the built-in readout. Open the console with the tilde key and type:

```
ACGA.Minimap.DebugDraw 1
```

WHAT YOU SEE	WHAT TO CHECK
The map is blank or grey	Is there an ACGA Map Capture in the level? Press its **Bake Background** button and look at the picture. If that is black, raise `Capture Height` — the camera is inside something.
The map is washed out or too dark	`Capture Exposure Bias` on the capture actor. Negative is darker. Re-bake after changing it.
No markers at all	Almost always no icon set. Check Project Settings → Plugins → ACGA Minimap → `Default Icon Set` is filled in.
One marker is missing	Its `Icon Id` is not in the set, its `Category` is not in that view's filter, `Start Visible` is off, or it is outside the capture box and the view's clamp mode is **None**.
Markers sit next to the terrain, not on it	The box was resized after the last picture was taken. Press **Bake Background**.
Clicking does nothing	`Selectable` on the marker *and* `Allow Selection` on the view — both are needed.
The full map will not open	`Full Map Enabled` in Project Settings.

Removing the plugin

Delete the markers and the capture actor, then untick the plugin. It does not replace any engine class and does not need anything in your GameMode or Player Controller, so a project that removes it still opens. If you pointed anything of your own at the plugin's assets, fix those references first, exactly as with any other asset.